

ISyE 3770 HW3

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Problem 3.3.6

Part A:

We observe the Cumulative Distribution Function to be defined as

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du$$

Therefore, we find:

$$F(x) = \begin{cases} 0, & x < 0 \\ 0.17, & 0 \leq x < 2 \\ 0.52, & 2 \leq x < 3 \\ 0.85, & 3 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$

Part B:

We find the Mean and Variance to be defined as

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

$$V(X) = E[X^2] - (E[X])^2$$

We will begin by solving for Mean:

$$E[X] = 0.17(0) + 0.35(2) + 0.33(3) + 0.15(4) = 2.29$$

Next, We will solve for Variance:

$$E[X^2] = 0.17(0^2) + 0.35(2^2) + 0.33(3^2) + 0.15(4^2) = 6.77$$

$$V(X) = 6.77 - (2.29)^2 = 6.77 - 5.2441 = 1.5259$$

Problem 3.4.3

As defined in notes, for a Discrete Random Variable, we have:

$$E[X] = \frac{a+b}{2}$$

$$V(X) = \frac{(b-a+1)^2 - 1}{12}$$

We easily find $a = 0$ and $b = 9$ as our upper and lower, inclusive, bounds. Therefore, we first calculate the Expected Value, utilizing the property of pulling out constants.

$$E[Y = 5X] = 5(E[X]) = 5\left(\frac{0+9}{2}\right) = 22.5$$

Next, we calculate Variance, utilizing the property of pulling out constants.

$$V(Y = 5X) = 5^2 V(X) = 5^2 \left(\frac{(9-0+1)^2 - 1}{12} \right) = 25(8.25) = 206.25$$

We find Standard Deviation to be defined as

$$SD(X) = \sqrt{Var(X)}$$

Therefore, we have

$$SD(Y) = \sqrt{Var(Y)} = \sqrt{206.25} = 14.37$$

Problem 3.5.13

Part A:

We recognize we are solving for $P(\text{number of passengers} \leq 120)$ and that this question models a Binomial Distribution.

$$f(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

To save time, we will calculate $1 - P(\text{number of passengers} > 120)$. Therefore, we have

$$P(X = 121) = \binom{125}{121} 0.1^{121} (1 - 0.1)^{125-121}$$

We will perform the above calculation for $X = 122$, $X = 123$, $X = 124$, $X = 125$.

```
choose(125, 121) * 0.9^121 * (1 - 0.9)^(125 - 121)+
choose(125, 122) * 0.9^122 * (1 - 0.9)^(125 - 122)+
choose(125, 123) * 0.9^123 * (1 - 0.9)^(125 - 123)+
choose(125, 124) * 0.9^124 * (1 - 0.9)^(125 - 124)+
choose(125, 125) * 0.9^125 * (1 - 0.9)^(125 - 125)
```

```
## [1] 0.003858595
```

Therefore, it is evident $1 - 0.00386 = 0.99614$ which is our final answer.

Part B:

We wish to find $P(\text{number of passengers} < 120)$. Alternatively, we wish to find $1 - P(\text{number of passengers} \geq 120)$. We remember we solved for a similar probability previously. Let's continue with the previous logic.

```
choose(125, 120) * 0.9^120 * (1 - 0.9)^(125 - 120)+
choose(125, 121) * 0.9^121 * (1 - 0.9)^(125 - 121)+
choose(125, 122) * 0.9^122 * (1 - 0.9)^(125 - 122)+
choose(125, 123) * 0.9^123 * (1 - 0.9)^(125 - 123)+
choose(125, 124) * 0.9^124 * (1 - 0.9)^(125 - 124)+
choose(125, 125) * 0.9^125 * (1 - 0.9)^(125 - 125)
```

```
## [1] 0.01143219
```

Therefore, it is evident $1 - 0.01143 = 0.98857$ which is our final answer.

Problem 3.6.6

Part A:

$$f(x) = p(X = x) = p(1 - p)^{x-1}$$

Since we know there is an 0.80 probability of defeating each enemy, we have a Geometric Distribution defined as:

$$f(x) = p(X = x) = 0.80(1 - 0.80)^{x-1}$$

Part B:

We observe the probability of a player reaching x level can be modeled by a simple Geometric Series:

$$P(X \geq x) = p^x = 0.8^x$$

Therefore, we find

$$P(X \geq 2) = 0.8^2 = 0.64$$

Part C:

We know the mean, or expected value of the first success, of a Geometric Distribution is defined as $E[X] = \frac{1}{p}$. Therefore, to find the first failure, we have $E[X] = \frac{1}{1-p}$.

$$E[X] = \frac{1}{1 - 0.8} = \frac{1}{0.2} = 5$$

Part D:

As defined in Part B, we have:

$$P(X \geq 4) = 0.8^4 = 0.4096$$

Problem 3.6.13

Part A:

We recognize this problem to be Negative binomial distribution with $r = 2$ and $x = 10$. Therefore, we have

$$f(x) = \binom{10-1}{2-1} 0.2^2 (1 - 0.2)^{10-2}$$

```
choose(10 - 1, 2 - 1) * 0.2^2 * (1 - 0.2)^(10 - 2)
```

```
## [1] 0.06039798
```

Part B:

We recognize this problem to be semi-identical to one covered in class, which uses a summation to calculate the probability.

$$P(X \leq x) = F(x) = \sum_{x=2}^x \binom{x-1}{2-1} (0.2)^2 (0.8)^{x-3}$$

```
pnbinom(q = 4 - 2, size = 2, prob = 0.2)
```

```
## [1] 0.1808
```

Part C:

The expected number of trials to get the third failure resembles a simple plug-and-chug from the Expected Value function for Negative Binomial Functions, defined as

$$E[X] = \frac{r}{1-p}$$

Therefore, we have $E[X] = \frac{3}{1-0.8} = \frac{3}{0.2} = 15$

Problem 3.7.6

Part A:

The probability of choosing one correct number is $\frac{6}{40}$. The probability of choosing two correct numbers is $\frac{6}{40} * \frac{5}{39}$ since there is one number less than before and the events are independent. As such we have:

```
(6/40)*(5/39)*(4/38)*(3/37)*(2/36)*(1/35)
```

```
## [1] 2.605266e-07
```

Therefore, we arrive at a final answer of 0.00000026 or 2.6×10^{-7} .

Part B:

We recognize the problem resembles a HYpergeometric Distribution, where $N = 40$ (# of items), $n = 6$ (sample), $k = 6$ (successes from population), $x = 5$ (successes from sample). Therefore, we have

$$f(x) = \frac{\binom{6}{5} \binom{40-6}{6-5}}{\binom{40}{6}}$$

```
choose(6,5)*choose(40-6,6-5)/choose(40,6)
```

```
## [1] 5.314742e-05
```

Therefore, we arrive at our answer of 0.0000531 or 5.31×10^{-6} .

Part C:

Using similar logic as Part C, we simply have $x = 4$ with the same other previously defined variables.

```
choose(6,4)*choose(40-6,6-4)/choose(40,6)
```

```
## [1] 0.002192331
```

Therefore, we arrive at our answer of 0.00219 or 2.19×10^{-3} .

Part D:

We find each week to represent a Bernoulli Trial, in which each week, there are two possible outcomes: Success or Failure.

As such, we define the Expected Value of the Bernoulli Distribution to be

$$E[X] = \sum_x xf(x) = p$$

We notice the probability of success was derived in Part A: 2.6×10^{-7} .

We wish to instead find $E[T]$, where T is the number of trials. Therefore, the expected number of trials until the first success is defined as

$$E[T] = \frac{1}{p}$$

Therefore, we arrive at our (estimated) answer of

$$E[T] \frac{1}{0.00000026}$$

1/0.0000002605266

[1] 3838380

Therefore, we have our final answer of 3838380 weeks until a player matches all six numbers in the state's sample.